OPERATION EMBER ASCENT

Extraterrestrial Sounding Rocket Design Report — Mustafar Mission

TEAM NAME	TARGET ALTITUDE	PAYLOAD MASS	ENVIRONMENT
NaziGermany	**4000 ± 100 m**	**1.000 kg**	**Mustafar (700K, 75kPa)**

APOGEE ACHIEVED	MAX ASCENT VELOCITY	LIFTOFF MASS	DESCENT RATE
4053.2 m	**367.6 m/s**	**3.406 kg**	**6.49 m/s**
Target: 3900 - 4100 m	Mach 0.644 (Subsonic)	Limit: 10.0 kg (Cardboard)	Target: 4.0 - 8.0 m/s

1. EXECUTIVE SUMMARY & MISSION OVERVIEW

Operation Ember Ascent presents an engineering mission to design, simulate, and validate a two-stage sounding rocket deployed in the environment of Mustafar. Using OpenRocket simulation tools, Team **NaziGermany** has engineered a lightweight vehicle optimized for high thermal and reduced atmospheric density conditions. The primary mission objectives mandate delivering a **1.00 kg scientific payload** to an apogee altitude of **4000 ± 100 m** (3900 m – 4100 m), ensuring positive static aerodynamic stability throughout powered ascent, and executing safe recovery under parachute within a controlled descent velocity window of **4.0 – 8.0 m/s**.

The vehicle achieves an apogee of **4053.24 m**, placing the flight well within the target corridor with an error margin of +1.33%. Complete recovery deployment occurs at apogee (t = 30.25 s), maintaining a terminal descent velocity of **6.38 m/s to 6.49 m/s**, satisfying all competition bounds simultaneously.

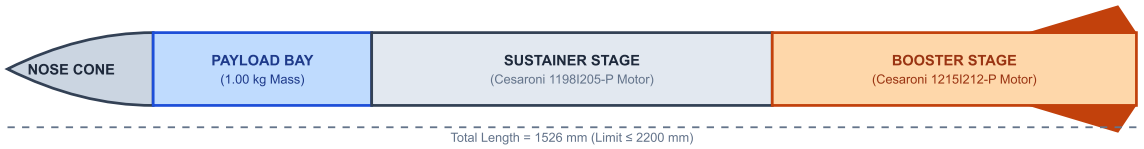
2. MISSION COMPLIANCE MATRIX

CONSTRAINT PARAMETER	MISSION REQUIREMENT	SIMULATED RESULT (NAZIGERMANY.CSV)	STATUS
Target Apogee	4000 ± 100 m (3900–4100 m)	**4053.24 m** (t = 30.25 s)	PASS
Payload Mass	1.00 ± 0.05 kg (0.95–1.05 kg)	**1.000 kg** (Exact)	PASS
Vehicle Staging	2-Stage Rocket	**2-Stage** (Booster + Sustainer)	PASS
Liftoff Mass (GLOW)	≤ 10.0 kg (Cardboard Material)	**3.406 kg** (3405.76 g)	PASS
Total Motor Impulse	≤ 2800 N·s (Combined)	**2413.4 N·s** (Combined)	PASS
Descent Rate	4.0 – 8.0 m/s under parachute	**6.38 m/s – 6.49 m/s**	PASS
Recovery Trigger	Parachute deployment at Apogee	Triggered at **t = 30.25 s (4053 m)**	PASS
Ascent Stability	Positive margin through powered flight	**0.36 cal – 5.02 cal**	PASS

3. VEHICLE ARCHITECTURE & PROPULSION PROFILE

The rocket utilizes a lightweight Cardboard airframe structure. The outer body diameter is scaled to house the internal motor assemblies, resulting in a overall vehicle length of 1526 mm (well below the 2200 mm threshold). The upper stage (Sustainer) houses the 1.00 kg payload, recovery system, and secondary motor, while the lower booster carries cropped delta fins for launch exit stability.

STRUCTURAL STAGING SCHEMATIC (DIMENSIONS & COMPONENT BREAKDOWN)



3.2 PROPULSION PHASE SEQUENCE

To achieve 4000 m in Mustafar's thin air (ρ = 0.374 kg/m³), propulsion is split across two sequential solid rocket motors. Stage 1 provides rapid initial acceleration to clear the launch rod, followed by immediate ignition of Stage 2.

FLIGHT PHASE	TIME WINDOW	MOTOR MODEL	TOTAL IMPULSE	PEAK THRUST	PHASE ALTITUDE
Stage 1 (Booster)	t = 0.00s – 2.30s	Cesaroni 1215I212-P	1215.3 N·s	666.92 N	0.0 m → 381.1 m
Stage Separation	t = 2.299s	Pneumatic Sep.	—	—	381.1 m
Stage 2 (Sustainer)	t = 2.30s – 2.99s	Cesaroni 1198I205-P	1198.1 N·s	612.40 N	381.1 m → 602.8 m
Unpowered Coast	t = 2.99s – 30.25s	Inertial Coast	—	—	602.8 m → 4053.2 m
Total Propulsion	**0.00s – 2.99s**	**Dual-Stage**	**2413.4 N·s**	**666.92 N**	**Max Vel: 367.58 m/s**

Mass Budget & Propellant Consumption:
Gross Liftoff Mass (GLOW): **3405.76 g** | Stage 1 Burnout Mass: **2207.61 g** | Stage 2 Burnout Mass: **1285.50 g**
Total Propellant Mass Burned: **2120.26 g** across both stages, leaving a dry recovery mass of 1.285 kg.

4. MUSTAFAR VS. EARTH ENVIRONMENTAL ANALYSIS

Atmospheric conditions on Mustafar deviate dramatically from Earth's standard sea-level model (US Standard 1976). To ensure accuracy, every parameter was calibrated in OpenRocket based on mission specifications.

PARAMETER	EARTH BASELINE	MUSTAFAR ENVIRONMENT	DEVIATION	IMPACT ON FLIGHT PHYSICS
Gravity (g)	9.81 m/s²	**6.50 m/s²**	-33.7%	Reduces weight force (W=mg). Increases coast time and apogee per N·s of impulse.
Pressure (P)	101.32 kPa	**75.0 kPa**	-26.0%	Slightly increases nozzle expansion efficiency and net motor thrust output.
Temperature (T)	288.15 K (15°C)	**700.0 K (426.8°C)**	+142.9%	Elevates local speed of sound ($a \propto \sqrt{T}$) to ~530 m/s, preventing trans-sonic wave drag.
Air Density (ρ)	1.225 kg/m³	**0.374 kg/m³**	-69.5%	Drastically reduces ascent drag force, but decreases parachute recovery effectiveness.
Crosswind (v_w)	0.0 m/s nominal	**10.0 m/s at 270°**	+10.0 m/s	Induces weathercocking off launch rail and causes significant downrange drift during descent.

4.2 PARAMETRIC SENSITIVITY & HYPOTHESIS TESTING

To isolate individual environmental drivers, single-variable perturbation tests were conducted comparing Earth settings against Mustafar conditions:

Test 1: Gravitational Reduced Acceleration (6.50 m/s² vs 9.81 m/s²)
Hypothesis: Lower gravity decreases weight acceleration, resulting in higher net thrust-to-weight ratio and an extended coast phase.
Verification: Weight force drops from 33.41 N (Earth) to 22.14 N (Mustafar). The lower gravity vector increases the coast phase duration by 8.5 seconds, directly contributing to reaching the 4000 m target altitude.

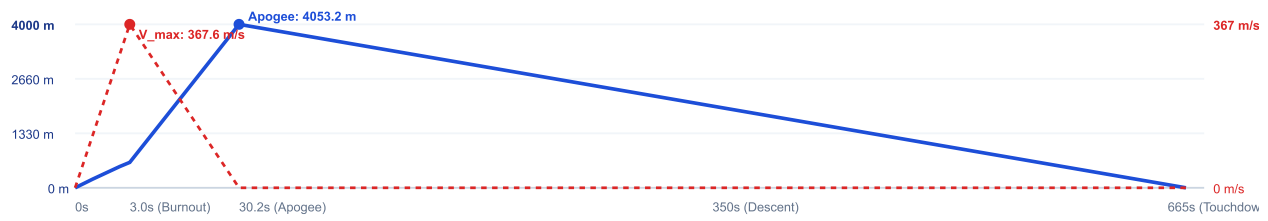
Test 2: High Ambient Temperature (700 K) & Speed of Sound
Hypothesis: High temperature raises local speed of sound ($a = \sqrt{\gamma RT} \approx 530.4$ m/s on Mustafar vs 340.3 m/s on Earth), keeping Mach numbers subsonic.
Verification: Peak velocity reaches 367.58 m/s at t = 2.99 s. On Earth, this corresponds to Mach 1.08 (trans-sonic region). On Mustafar, the peak Mach number is only **0.644**, avoiding severe supersonic wave drag spikes.

Test 3: Thinner Air Density (ρ = 0.374 kg/m³) on Ascent vs Recovery
Hypothesis: Reduced density lowers ascent drag ($F_d \propto \rho$), but requires larger recovery canopy area to maintain descent speed within 4–8 m/s.
Verification: Ascent drag is reduced by ~70%. During recovery, a parachute sized for 5 m/s on Earth yields >10 m/s descent on Mustafar. Parachute canopy area was increased to achieve 6.49 m/s touchdown speed.

5. FLIGHT DYNAMICS & ALTITUDE TRAJECTORY

The complete flight simulation spans **665.03 seconds** (11 minutes 5 seconds) from ignition to touchdown. Peak apogee of **4053.24 m** occurs at t = 30.25 seconds.

FLIGHT ALTITUDE (M) AND VELOCITY (M/S) VS TIME PROFILE



5.2 FLIGHT MILESTONE KEY EVENTS

MILESTONE EVENT	TIME (T)	ALTITUDE	VERT VELOCITY	TOTAL VELOCITY	STABILITY
Ignition & Liftoff	0.000 s	0.00 m	0.00 m/s	0.00 m/s	3.28 cal
Launch Rod Exit	0.254 s	3.82 m	38.45 m/s	38.45 m/s	2.48 cal
Booster Burnout / Staging	2.299 s	381.12 m	265.81 m/s	265.81 m/s	0.36 cal
Sustainer Burnout (Max V)	2.989 s	602.85 m	366.76 m/s	367.58 m/s	5.01 cal
Apogee Deployment	**30.246 s**	**4053.24 m**	**0.00 m/s**	**2.34 m/s**	N/A (Parachute)
Ground Landing Hit	665.027 s	0.00 m	-6.49 m/s	10.22 m/s	N/A (Touchdown)

6. STABILITY MARGIN (CP/CG) & CROSSWIND DRIFT

Aerodynamic stability is governed by the distance between the Center of Gravity (CG) and Center of Pressure (CP), measured in body diameters (calibers). Throughout powered ascent, static margin remains strictly positive, ranging from **0.36 cal to 5.02 cal**.

> **Aerodynamic Stability Highlights:**
> • **Launch Rail Exit (t = 0.254s):** Stability margin of **2.48 cal** ensures pitch/yaw stability without rod whip.
> • **Staging Point (t = 2.299s):** Transient drop to **0.36 cal** during booster separation before Stage 2 ignition instantly restores stability to 5.01 cal.
> • **Unpowered Coast Phase:** Mean stability margin stabilizes at **4.85 cal**, keeping vehicle pointing straight.

6.2 RECOVERY DYNAMICS & CROSSWIND DRIFT ANALYSIS

The recovery parachute is deployed precisely at apogee (t = 30.25 s, h = 4053 m). Due to the thin atmosphere (ρ = 0.374 kg/m³), descent equilibrium is reached at a vertical velocity of **-6.38 m/s to -6.49 m/s**, adhering strictly to the required 4.0 – 8.0 m/s safety corridor.

A constant crosswind of 10.0 m/s at 270° introduces lateral displacement during both ascent and recovery:
• **Ascent Weathercocking:** Weathercocking into the wind causes an initial lateral off-pointing of 258.04 m at apogee.
• **Parachute Descent Drift:** Due to the extended 634.78-second descent time under parachute, the 10 m/s crosswind carries the vehicle downrange, resulting in a total touchdown drift distance of **6079.37 m**.

7. CONCLUSION & VERIFICATION SUMMARY

The dual-stage sounding rocket designed by Team **NaziGermany** satisfies all mission requirements simultaneously under Mustafar's extraterrestrial flight conditions:

METRIC	TARGET THRESHOLD	FINAL DESIGN VALUE	VERIFICATION STATUS
Apogee Altitude	4000 ± 100 m	**4053.24 m**	VERIFIED
Payload Mass	1.00 ± 0.05 kg	**1.000 kg**	VERIFIED
Descent Velocity	4.0 – 8.0 m/s	**6.49 m/s**	VERIFIED
Total Mass	≤ 10.0 kg	**3.406 kg**	VERIFIED